1. $\alpha:(-\varepsilon,\varepsilon)\to M$ $\quad \alpha=\alpha_1+\alpha_2$

则 $\phi\circ\alpha=\phi(\alpha_1+\alpha_2)=\phi\circ\alpha_1+\phi\circ\alpha_2$

$$\frac{d(\phi\circ\alpha)}{dt}(0)=\frac{d(\phi\circ\alpha_1)+d(\phi\circ\alpha_2)}{dt}=[\alpha_1]+[\alpha_2]\in T_xM$$

$$\frac{d(\phi\circ c\alpha)}{dt}(0)=\frac{cd(\phi\circ\alpha_1)}{dt}=c[\alpha_1]\in T_xM$$

2 M_1 平面参数化 $r(u,v)=(u,v,1)$ $\quad E=G=1,\ F=0 \quad I_1=du^2+dv^2$

M_2 柱面参数化 $r(\theta,z)=(\cos\theta,\sin\theta,h)$ $\quad \theta\in[0,2\pi),\ h\in[-2,1]$ $\quad \tilde{E}=\tilde{G}=1,\ \tilde{F}=0,\ I_2=d\theta^2+dh^2$

$\psi: M_1\to M_2$

$(u,v,1)\mapsto(\cos u,\sin u,v)$ 光滑

且 $\psi^*I_2=I_1$

$\psi^*I_2=I_1$

故 M_1 M_2 等距

3. 由 Nash-Kuiper Th. 存在 C^1 等距嵌入 $\mathbb{R}^3$

则存在 C^0 等距嵌入 $\mathbb{R}^3$

$\forall(s,t)\in[0,1]\times[0,1]$ $\quad (s,0)\sim(s,1),\quad (0,t)\sim(1,t)$

$$f(s,t)=\begin{cases}(0,4s,t) & 0\le s<\frac{1}{4}\\ (4s-1,1,t) & \frac{1}{4}\le s<\frac{2}{4}\\ (1,3-4s,t) & \frac{2}{4}\le s<\frac{3}{4}\\ (4-4s,0,t) & \frac{3}{4}\le s\le 1\end{cases}$$

f 连续保距.

4. 1° 显然 $\mathbb{R}P^2$ 是紧 Hausdorff 第二可数的拓扑空间

2° $U_1=\{(x_1,x_2,x_3)\in S^2 \mid x_1>0\}$

$U_2=\{(x_1,x_2,x_3)\in S^2 \mid x_2>0\}$

$U_3=\{(x_1,x_2,x_3)\in S^2 \mid x_3>0\}$

$\pi: S^2\to\mathbb{R}P^2$

$x\mapsto[x]$

令 $V_i=\pi(U_i)$

$\phi_1: V_1\to\mathbb{R}^2$

$[x_1,x_2,x_3]\mapsto\left(\frac{x_2}{x_1},\frac{x_3}{x_1}\right)$

$\phi_2: V_2\to\mathbb{R}^2$

$[x_1, x_2, x_3] \mapsto (\frac{x_1}{x_2}, \frac{x_1}{x_2})$

$\phi_3: V_3 \to \mathbb{R}^2$

$[x_1\ x_2\ x_3] \mapsto (\frac{x_1}{x_3}, \frac{x_2}{x_3})$

下证 $\forall i,j=1,2,3$ $\phi_j \circ \phi_i^{-1}$ 光滑

以 $\phi_1 \circ \phi_2^{-1}$ 为例: $\phi_2^{-1}(u,v) = [u, 1, v] \sim [\frac{u}{\sqrt{1+u^2+v^2}}, \frac{1}{\sqrt{1+u^2+v^2}}, \frac{v}{\sqrt{1+u^2+v^2}}]$

$\therefore \phi_1 \circ \phi_2^{-1}(u,v) = (\frac{1}{u}, \frac{v}{u})$.

$\therefore \mathbb{R}P^2$ 是二维光滑流形.

$\forall X, Y \in T_p S^2$. $g_{S^2}(X,Y) = \langle X, Y\rangle|_{\mathbb{R}^2}$

$\sigma: S^2 \to S^2$

$x \mapsto -x$

则 $d\sigma_p(X) = -X \in T_{\sigma p} S^2$

$\sigma^* g_{S^2}(X,Y) = g_{S^2}(d\sigma(X), d\sigma(Y))$

$= \langle -X, -Y\rangle$

$= \langle X, Y\rangle$

$= g_{S^2}(X,Y)$

$\therefore \sigma$ 等距

定义: $\forall [p] \in \mathbb{R}P^2$. $X, Y \in T_{[p]}(\mathbb{R}P^2)$

$g_{[p]}(X,Y) = g_{S^2}(\tilde{X}, \tilde{Y})$, 其中 $d\pi_p(\tilde{X}) = X$ $d\pi_p(\tilde{Y}) = Y$

良定义

5. $r(u,v) = (u, v, u^2 - \frac{v^2}{4})$

$r_u = (1, 0, 2u)$

$r_v = (0, 1, -\frac{v}{2})$

$r_{uu} = (0, 0, 2)$

$r_{uv} = (0, 0, 0)$

$r_{vv} = (0, 0, -\frac{1}{2})$

$E = 1 + 4u^2$

$F = -uv$

$G = 1 + \frac{v^2}{4}$

$\vec{N} = \frac{(-2u, \frac{v}{2}, 1)}{\sqrt{1+4u^2+\frac{v^2}{4}}}$

$L = \frac{2}{\sqrt{1+4u^2+\frac{v^2}{4}}}$

$M = 0$

$N = \frac{-\frac{1}{2}}{\sqrt{1+4u^2+\frac{v^2}{4}}}$

$$\Gamma^1_{11} = \frac{4u}{1+4u^2+\frac{v^2}{4}} \qquad \Gamma^1_{12} = 0 \qquad \Gamma^1_{22} = \frac{-u}{1+4u^2+\frac{v^2}{4}}$$

$$\Gamma^2_{11} = \frac{-v}{1+4u^2+\frac{v^2}{4}} \qquad \Gamma^2_{12} = 0 \qquad \Gamma^2_{22} = \frac{\frac{v}{4}}{1+4u^2+\frac{v^2}{4}}$$

$$N_u = \frac{(2+\frac{v^2}{2},\ 2uv,\ 4u)}{(1+4u^2+\frac{v^2}{4})^{\frac{3}{2}}} \qquad L_v - M_u = -\frac{v}{2(1+4u^2+\frac{v^2}{4})^{\frac{3}{2}}}$$

$$N_v = \frac{(uv,\ 1+4u^2,\ -\frac{1}{2}v)}{2(1+4u^2+\frac{v^2}{4})} \qquad M_v - N_u = -\frac{2u}{(1+4u^2+\frac{v^2}{4})^{\frac{3}{2}}}$$